

ATPG session 3

Case 8 :- Seq depth + Unclassified faults

↳ There are some undriven nets in the netlist.

Inform it to the fr team.

If those nets are driven properly, then you will be able to cover those faults.

Eg ①:

AU.SEQ :-

Assume a fault : SA0
Identify opp value : 1

Tool will try to control the fault site to 1
Prop the value at fault site to an obs pt X ✓

```
ANALYSIS> analyze_fault /U3/Y -stuck_at 0 -D
// -----
// Fault analysis for /U3 (373) output Y stuck at 0
// -----
// Current fault classification = AU (atpg_untestable)
// Fault site sequential depth: Control_0 = 0, Control_1 = 0, Observe = 255.
// Analyze if the fault can be activated ...
// The fault site was set to 1 successfully (data in parallel_pattern 0).
// Pattern type: Clock_sequential (cycles = 1).
// Search potential observation point...
// Fault is blocked at /U11 (522) due to /U16 (458) unable to set to 0.
// Tied gate source: X /addr[1] (103)
// Fault is blocked at /U10 (529) due to /U15 (523) unable to set to 0.
// Tied gate source: X /addr[0] (104)
// Fault is blocked at /U9 (524) due to /U14 (460) unable to set to 0.
// Tied gate source: X /addr[0] (104)
// Fault is blocked at /U8 (530) due to /U13 (372) unable to set to 0.
// Tied gate source: X /addr[1] (103)
```

Controlling is not a problem.

Reason why the value at fault site is not able to prop to an obs pt.

Prop to an obs pt is the problem

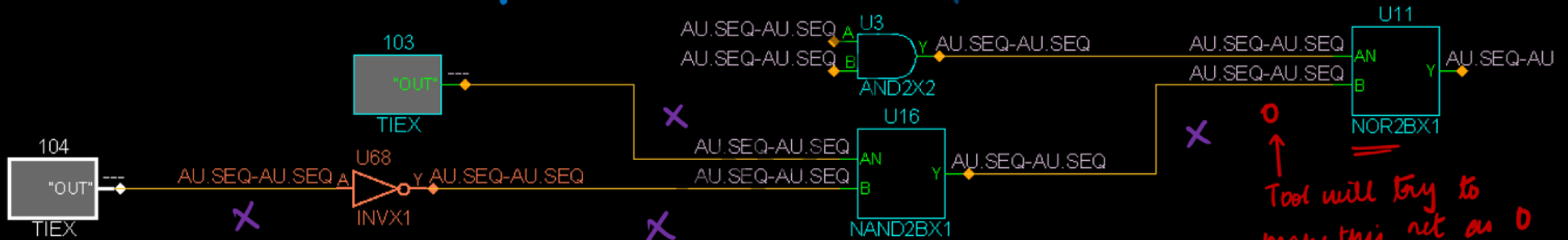
Analysis summary:

The fault site cannot propagate to any observation point.

U3/y SAB

tool was able to control the fault site to 1

value at fault site has to be prop to an obs pt



Tool will try to make this net as 0 to prop the value at the fault site to an obs pt.

But tool can't bring 0 here as there are some undriven nets in netlist

NOR Gate TF

i1	i2	y
0	0	1
0	1	0
1	0	0
1	1	0

only when i/p = 0, o/p depends on other i/p

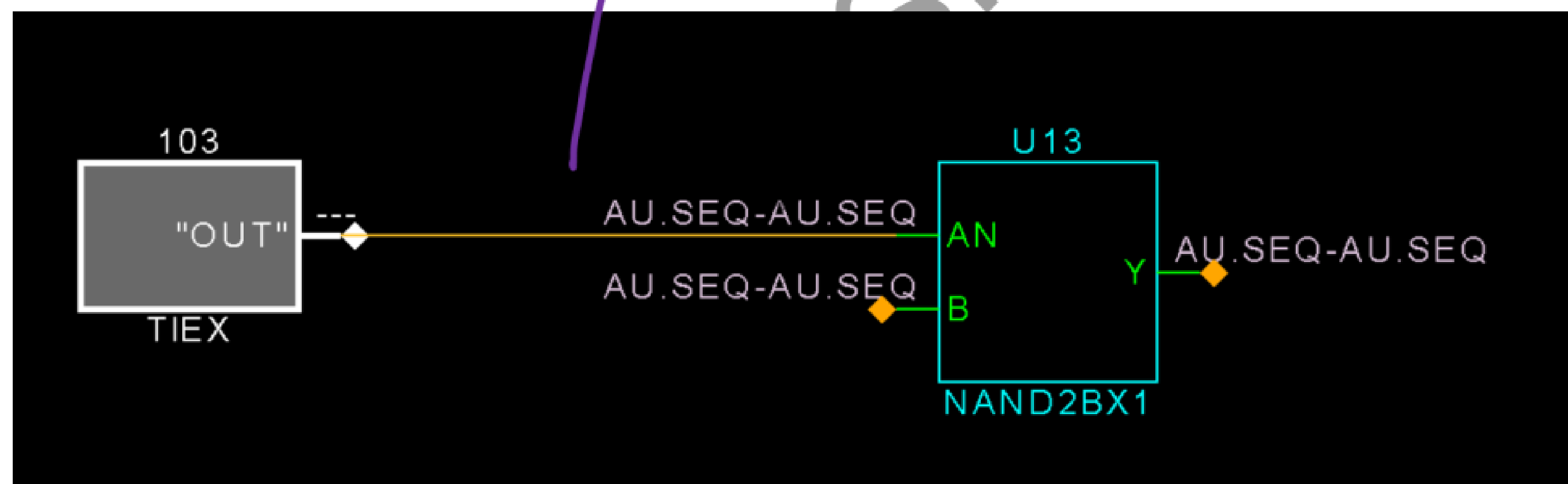
Eg 2 :

Assume a fault: SA0
identify opp value: 1
Tool will try to control the fault site to 1 X

```
ANALYSIS> analyze_fault /U13/AN -stuck_at 0 -D
// -----
// Fault analysis for /U13 (369) input AN (0) stuck at 0
// -----
// Current fault classification = AU (atpg_untestable)
// Fault site sequential depth: Control_0 = 255, Control_1 = 255, Observe = 0.
// Fault site cannot be set to 0 or 1 due to constrained logic.
// Tied gate source: X /addr[1] (103)
```

→ Controlling is a problem.

→ U13's AN is connected to a wire which is undriven
∴ tool is not able to control AN to an appropriate value.



Eg③:

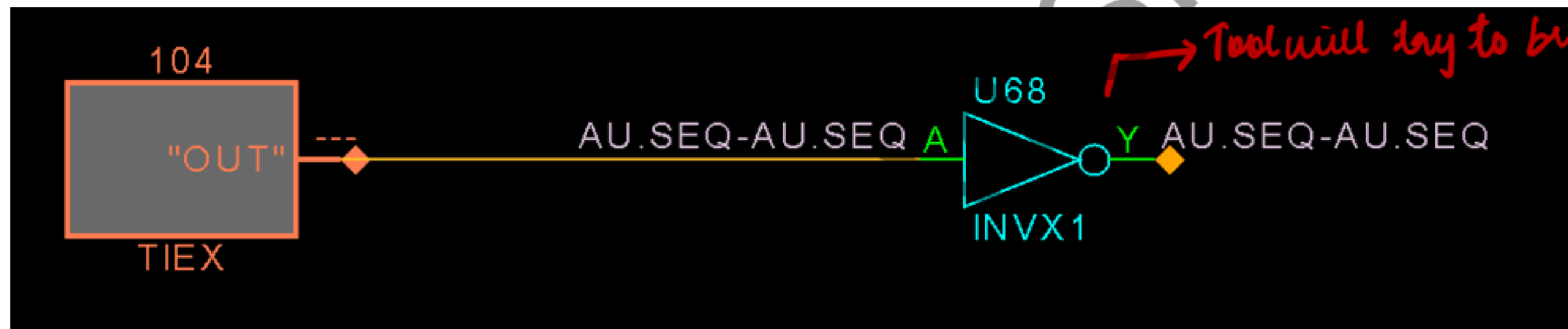
Assume a fault: SA0
Identify opp value: 1

ANALYSIS> analyze_fault /U68/Y -stuck_at 0 -D

Tool will try to control the fault site to 1.X

```
// -----  
// Fault analysis for /U68 (370) output Y stuck at 0  
// -----  
// Current fault classification = AU (atpg_untestable)  
// Fault site sequential depth: Control_0 = 255, Control_1 = 255, Observe = 0.  
// Fault site cannot be set to 0 or 1 due to constrained logic.  
// Tied gate source: X /addr[0] (104)
```

↳ Controlling is the problem.

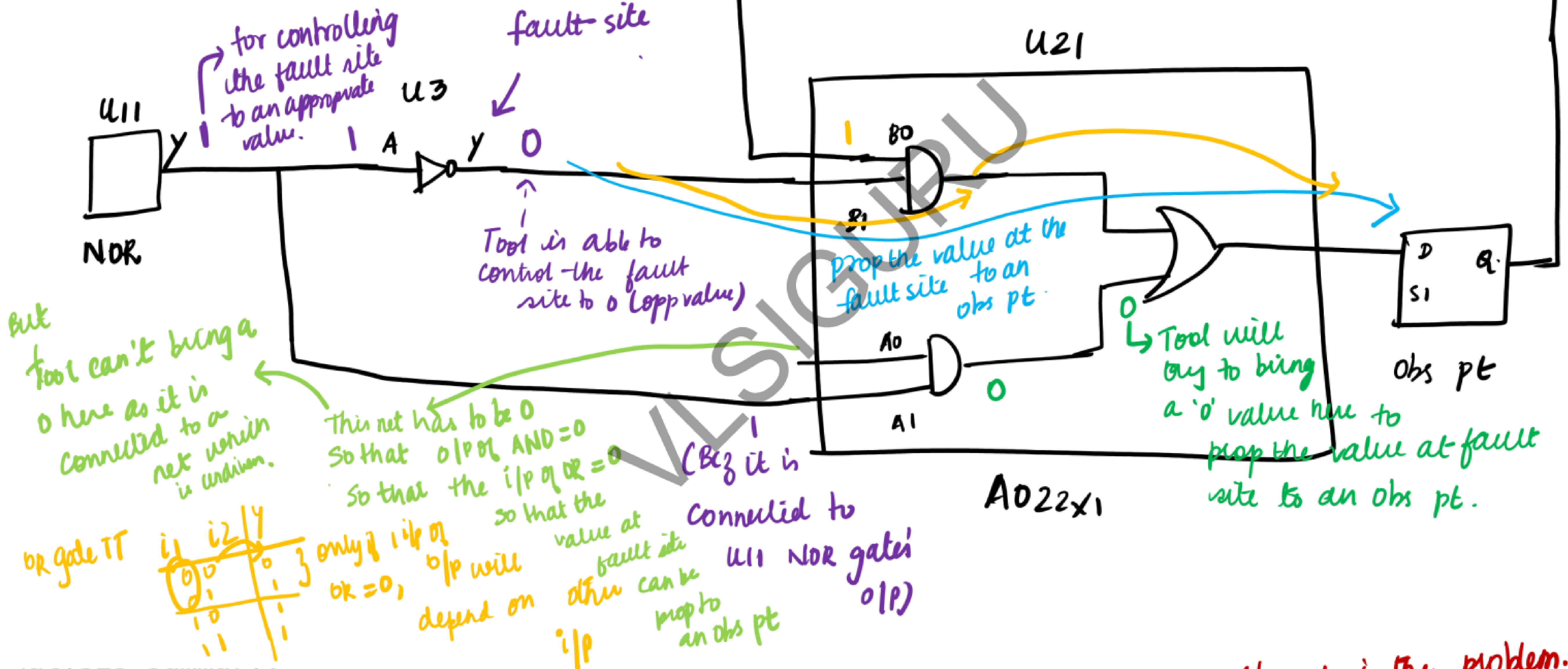


Tool will try to bring 1 at the o/p of NOT gate.
↳ Tool will try to make the i/p of NOT gate as 0.
But it can't make the i/p as 0 as it is connected to a net which is undriven.

AU.Unclassified:

The fault site was set to 0 successfully

Eg ①: Assume a fault: SA1
 U3/y SA1 identify opp value: 0 ✓



The fault site cannot propagate to any observation point. ⇒ prop to an obs pt is the problem. why? see above

Case 9:

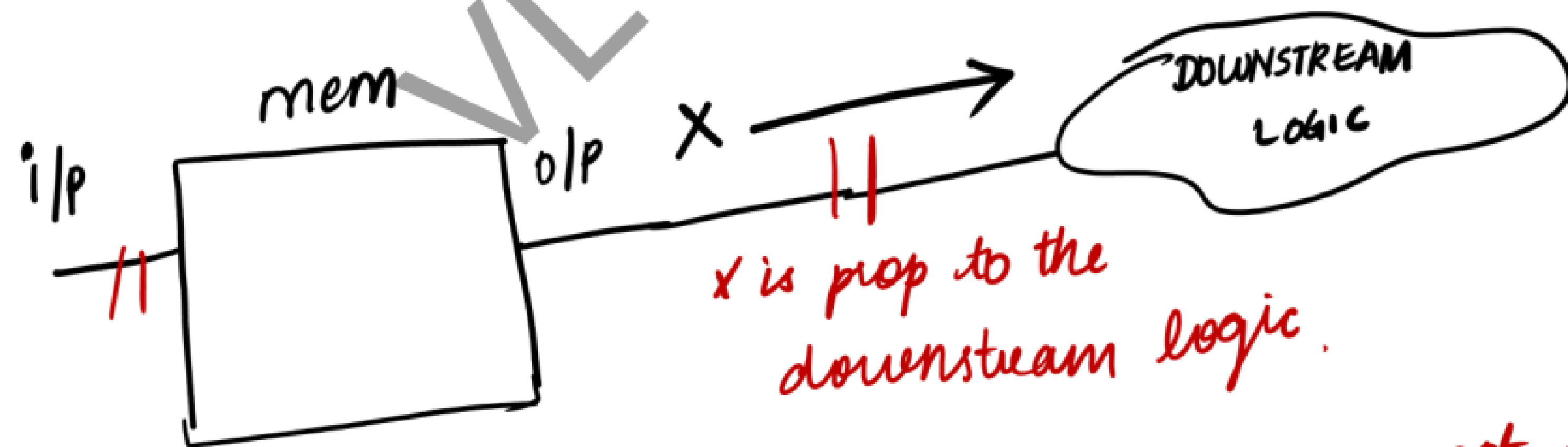
Memories in the design.

We use separate MBIST technique to test the memories

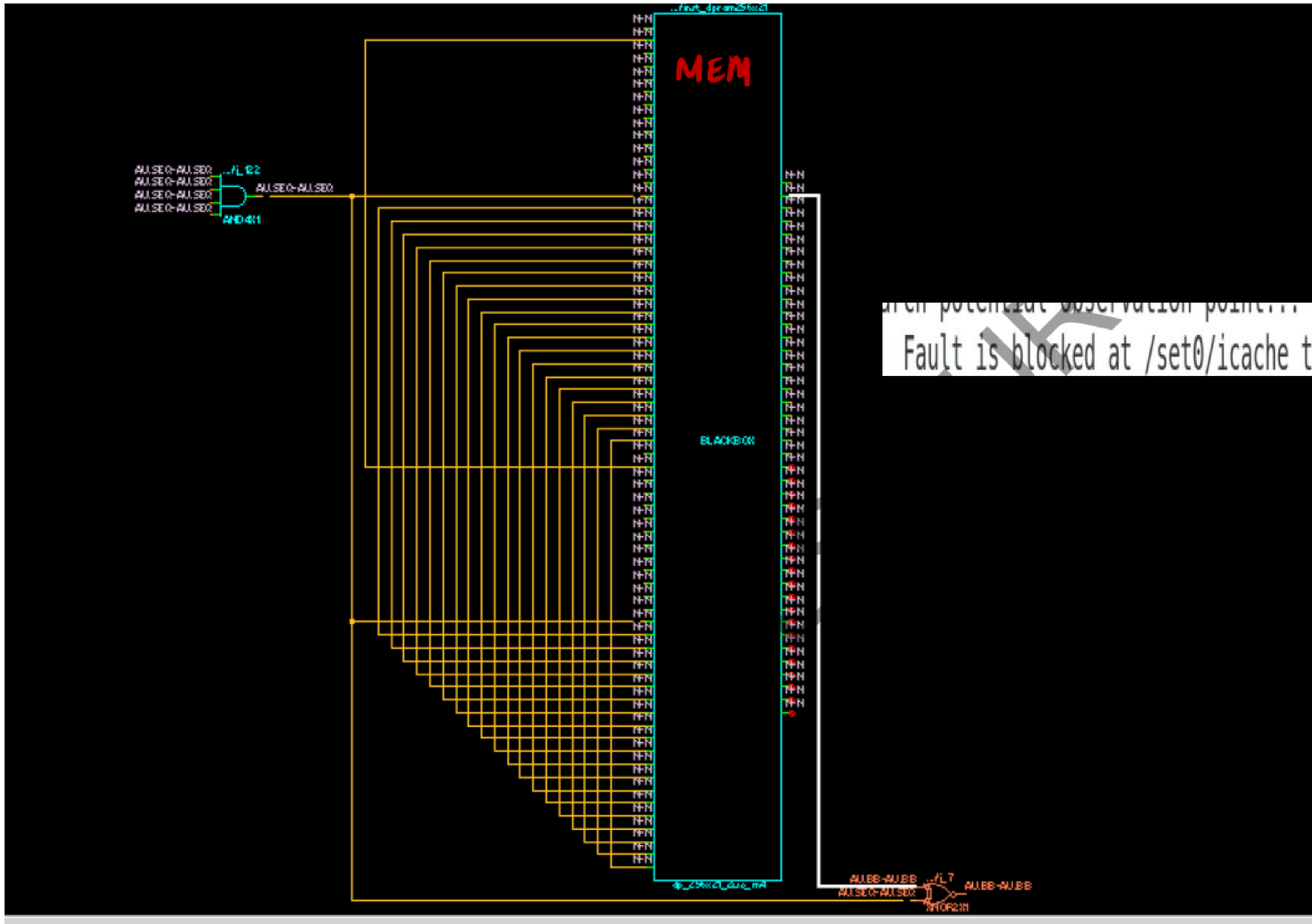
ATPG tool considers the o/p of memories as X.

So some faults will be placed in AU.SEB

We have also ^(Black Box) BB the memory. So some faults are placed in AU.BB class of faults.



// → faults are not detected



...potential observation point...

Fault is blocked at /set0/icache tag/inst dpram256x21 (337) due to tieX behavior.

Soln:- Bypass the mem to stop the prop of x to the downstream logic:

• ted-mem-lib

↳ Transparent Mode: SyncMux

↳ Tool will automatically add the logic to bypass the memory.

SFF

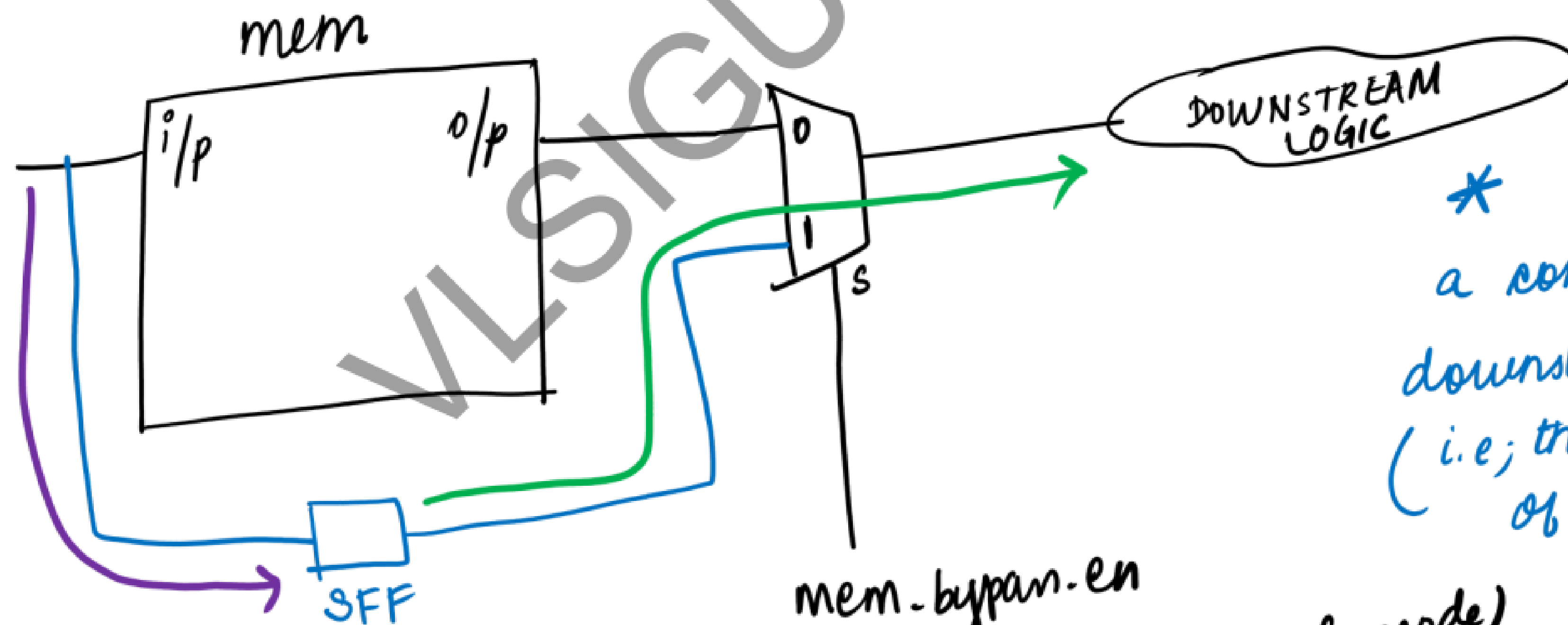
□ :-

It will act as an Obs pt for the fault sites at the i/p of the mem.

(i.e., the value at the i/p of mem will be Cap into that ff.)

[depicted by violet line]

See *



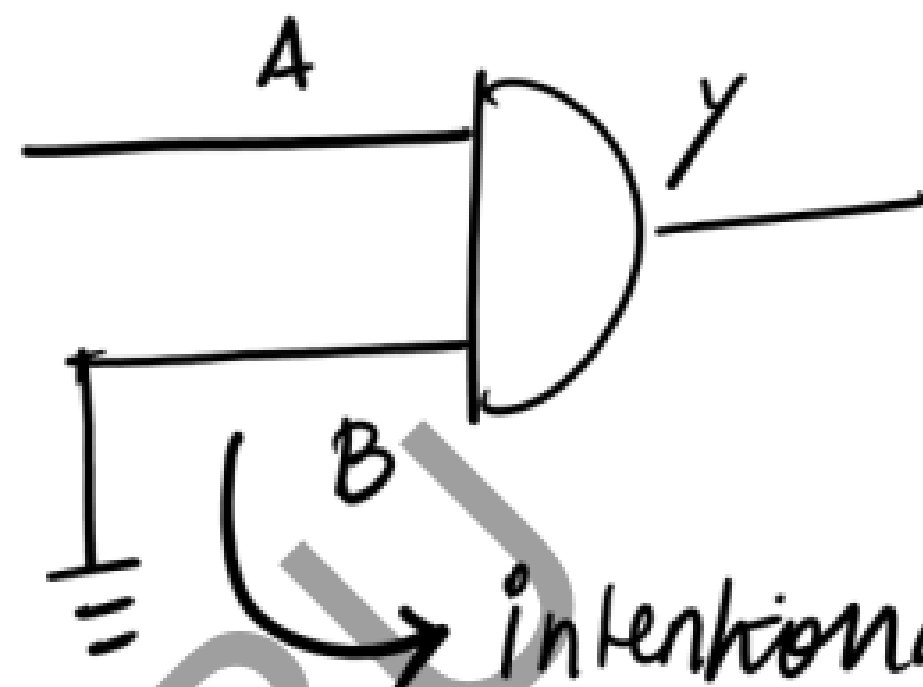
mem-bypass-en
= 0 (During MBIST or fn mode)
= 1 (otherwise)

* It will act as a control ff for the downstream logic.
(i.e.; the last shifted value of the ff will control the downstream logic)

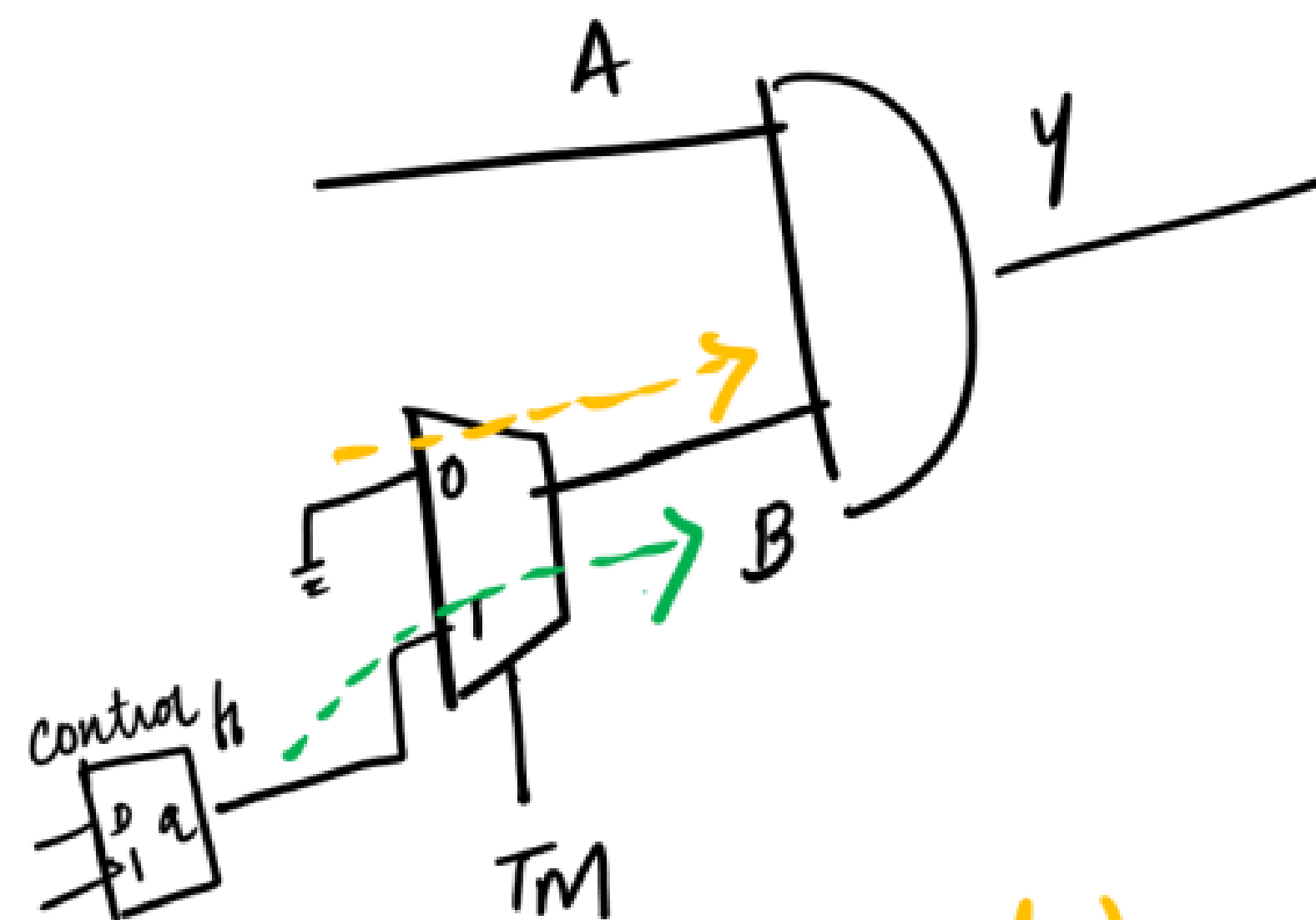
[depicted by green colour line]

NOTE :

Tied :-

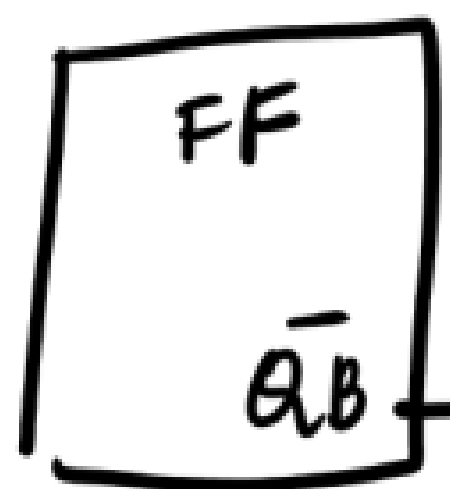


intentionally done by the designer
SAO on this net $\Rightarrow$ T1



= 0 (fn mode)
= 1 (CDFI mode)

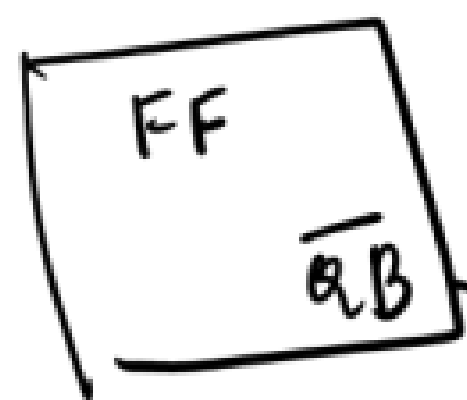
Covering untestable faults :



unused (UU)

$\hookrightarrow$ This is not connected
anywhere and hence does
not go to any obs pt.

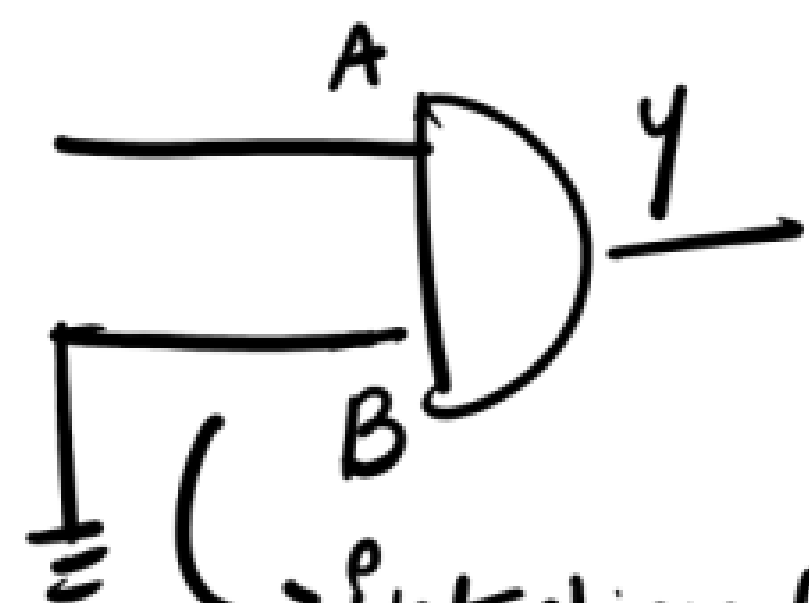
You can cover the faults by adding an
obs pt



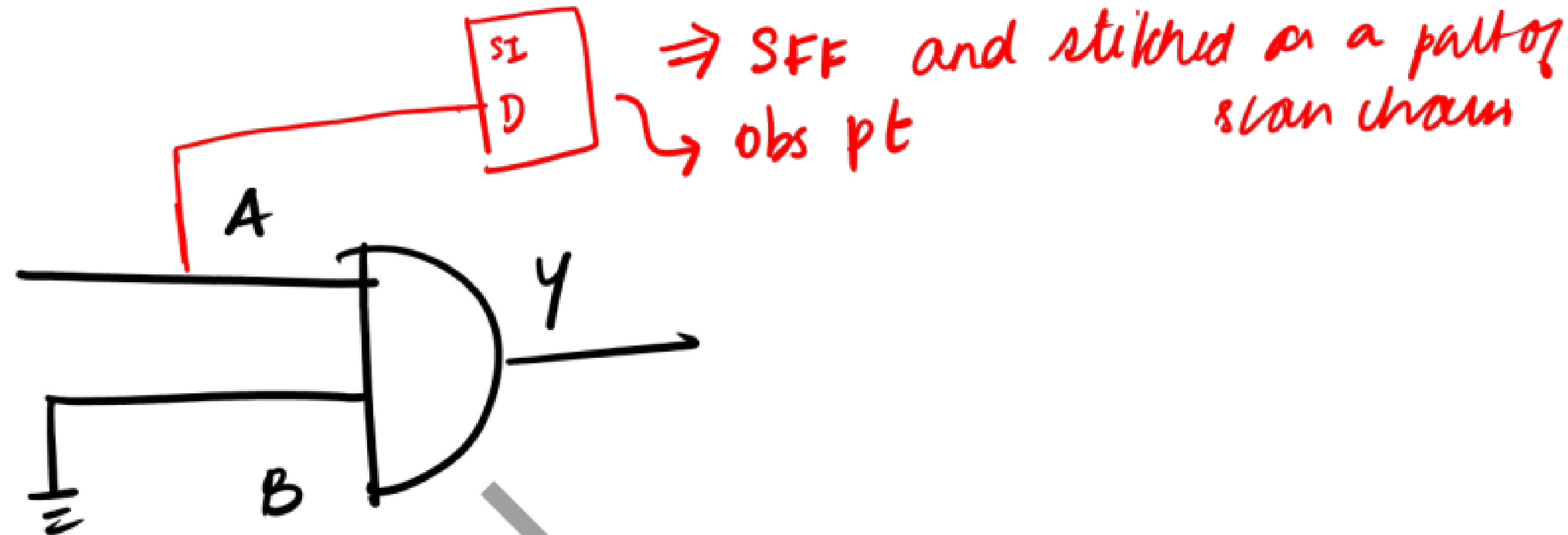
FF $\rightarrow$ Scan ff and
stitched as a part
of scan chain.

obs pt

Blocked :



Intentionally done by the designer



Similarly for redundant fault class, we can add obs pt.

Faults on net A is blocked

↳ can't be propagated to an obs pt.

Why? Since B is tied to Gnd, $Y=0$ irrespective of the value of A.

But we usually don't cover the uncontrollable faults because if suppose any of the faults in this class occurs after manufacturing and even if it is not detected, it will not cause any fn failures. (For understanding, please refer to thy notes)

So in real time, we don't spend time and manpower to cover the uncontrollable faults.

∴ In real time we will only focus on improving the Test cov.